Noninvasive EEG-Based Intelligent Mobile Robots: A Systematic Review

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***Abstract*— Brain-controlled mobile robotics can provide restoration of mobility for individuals with severe physical disabilities and empower healthy people with a broader reachable range in particular environments, which have been flourished over the past twenty years. This paper conducts a systematic state-of-the-art overview of noninvasive EEG-based intelligent mobile robots. We first present the general architecture and basic concepts, typical system types, and main research efforts on the whole-system design. Then, relevant key techniques associated with the brain-machine interfaces (BMIs), control strategies, and robot intelligence are reviewed to elucidate the research progress of the overall system. System performance evaluation is critical and complicated, here we summarize the conditions of the recruited participants, the experimental protocol, tasks and environments, with an emphasis on evaluation metrics regarding BMI performance, navigation performance, system robustness, and the user. We further highlight the remaining challenges and the potential research directions of future work. This study with informative outline is envisioned to enhance current understanding and suggest the future perspectives on EEG-based mobile robotic devices.**

***Note to Practitioners*—The brain-computer interfaces (BCIs) can provide a novel and feasible way to convey the users' intentions to the terminal devices through the brain signals, which has exhibited emerging prospects in both civil and military applications thus having attracted increasing interests in researchers, industries, and government. The integration of noninvasive electroencephalogram (EEG)-based BCIs and expanded wheeled robotics have potentials to improve the human daily life with enhanced level of mobility, exhibiting significant socioeconomic impacts. This paper provides a systematic review of the EEG-based intelligent mobile robots from the aspects of key milestones, key techniques, and performance evaluation. To bring such the next generation assistive products from the laboratory to real applications, this paper further highlights the remaining challenges and shed light on the potential future work.**

***Index Terms*— Electroencephalography, intelligent robots, brain-controlled robotic systems, human-machine interaction, collaborative control, performance evaluation.**

I. Introduction

MOBILITY refers to the capacity to move flexibly in a given direction and can be provided by assistive mobile robotics with various structures such as wheeled, legged, crawler, hybrid, and so on [1]. Among which, wheeled robots have been extensively investigated to meet the mobility and transportation demands of humans in various environments due to the characteristics of fast movement, high traction, and simple wheel configuration [2], [3]. Here and in the remainder of this paper, wheeled robotics refer to wheelchairs, wheeled mobile robots, and vehicles equipped with wheels.

Traditionally, a user controls wheeled robots via the entity mechanical devices (e.g., the mouse, keyboard, button, handle, and joystick). Yet the requirement of operator's body motions limit the application to those with motor impairment. Researchers have developed special interfaces for these people, such as the voice recognition system, eye tracking system, and sip-and-puff system [4], [5]. However, these interfaces still rely on the user's neuromuscular tissues and are not suitable for people suffering from spinal cord injuries (SCIs), multiple sclerosis (MS), amyotrophic lateral sclerosis (ALS), and strokes. On the contrary, the brain-computer interface (BCI) technique, also known as brain-machine interface (BMI), bridges the human brain and external world via a non-muscular communication channel [6], and allows wider application scenarios. It has already been successfully used in cursor control, word spelling, web browsing, and other assistive systems [7], [8]. To develop more powerful assistive devices, mobile robotics employing the BCI technique appeared, which are called brain-controlled mobile robotics.

The basic idea of a brain-controlled robotic system is to interpret user intents based on the recorded neurophysiological brain signals and transform them into control actions to further actuate the external robots. Considering the health risks and ethical concerns associated with invasive techniques using implanted electrodes, relevant studies in brain-robot research have primarily focused on non-invasive brain activity

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monitoring that does not require surgery, such as modalities of electroencephalogram (EEG), magnetoencephalography (MEG), near-infrared reflectance spectroscopy (NIRS), and functional magnetic resonance imaging (fMRI) [7]. In particular, EEG has been the most commonly used recording method because of its great temporal resolution, nonwound, easy operation, relatively low cost, and portability [9], [10].

Millán et al. investigated the first EEG-based mobile robot in 2004, and demonstrated brain control of a miniature mobile robot [11], pioneering the field [12], [13], [14], [15]. Later, various intelligent approaches in user intention decoding were investigated, and control strategies (how to blend robot intelligence with human output) accompanied by advanced sensing technologies were emerged throughout the first ten years. To date, the integration of BCI/BMI and expanded wheeled robotics have thrived to provide users with enhanced assistive mobility and transportability. Such a combination has demonstrated its feasibility and effectiveness in several scenarios. At an early stage of EEG-based robotic research, brain-controlled wheelchairs (BCWs) were developed to assist handicapped individuals with severe neuromuscular problems in regaining self-independence and improving their quality of life [16], [17]. Recent brain-controlled vehicles (BCVs), to go a step further, may allow people with locked-in disorders to drive, and provide an alternative car-driving strategy to better assist healthy people [18], [19], [20]. Furthermore, in some specialized scenarios such as demining, search and rescue, surveillance, and exploration, enhanced reachable range through brain teleoperation (i.e., brain-controlled teleoperated mobile robots, BCTMs) can provide assistance in performing complicated tasks in remote locations [21]. Nevertheless, these brain-inspired ecosystems are still in their infancy.

A. Related Work

Several reviews have outlined the past progresses of EEG-based wheeled robotics. For instance, Bi et al. published a comprehensive review of EEG-based mobile robots in [22], summarizing the progresses of such systems up to 2013, mainly focusing on the key techniques and evaluation issues. Another review of BCI-powered wheelchairs was presented in [23], where the variations in brain signals were mainly clarified for gathered 34 papers. Similarly, distinct EEG signal processing algorithms were summarized and reported in detail [24], [25], [26], [27]. To better comprehend the nearest ten-year field development, Hekmatmanesh et al. [28] conducted an extensive literature review including both terrestrial and aerial BCVs. However, the details of the system evaluation were not disclosed, in particular the information on system performance metrics and participants. In 2020, Tonin released an exciting evaluation of BMI-based robotic devices, deeply emphasized the problems of mutual learning interactions among the user, BMI system, and robots [29]. Unfortunately, the latest research on one of the most representative mind-controlled wheeled robots, that is, BCVs, has not been involved. Fu et al. reported a quantitative analysis of the latest EEG-based devices [30], with the typical studies within five years (from 2017 to 2021) synthesized and analyzed. Moreover, a latest review has been conducted in [31] to examine the research state of EEG-driven wheelchairs and emphasize the need to enhance the practicality of BCI systems for disabled individuals.

These above published studies have provided the community with a fundamental understanding and general knowledge of EEG-based intelligent mobile robots. Nevertheless, several issues have not been further clearly elucidated. One of the most prominent shortcomings is that most reviews aforementioned focus more on the research progress of BCI itself rather than on the whole brain-controlled systems, thus the relevant advances with the controlled terminal components are often ignored. It also leads to a second problem: the lack of a truly systematic summary in terms of the targeted applications, specified techniques, and system evaluation. Some existing studies have not been properly categorized to clarify the advancement of each research focus, and those new, meaningful, and cutting-edge investigations on the emerging research hotspots have not been included. For instance, several novel BCI paradigms have attracted increasing interests such as the steady-state somatosensory evoked potential (SSSEP) [32], applied in BCWs [33], and tactually-evoked event-related potentials (ERP) [34], effectively utilized for virtual wheelchair control [35]. The brain control of multiple coordinated robots through a BMI has also been achieved [36], [37], [38], considerably advancing the research of a single robot and expanding the control with diverse tasks. However, there is a lack of a thorough overview including these advanced studies.

B. Contribution and Overview

This article aims to provide a broad and systematic overview of noninvasive EEG-based intelligent mobile robotics over the past twenty years. We restricted the focused terminal devices to ground wheeled robots (i.e., wheelchairs, wheeled mobile robots, and vehicles) due to their ability to provide relevance in meeting mobility and movement needs while possessing the typical features of dynamic system control, such as the rapid timing requirement, complex control sequences with multiple commands, and planar motion within two-dimensional (2D) space while exhibiting the inherent mechanical characteristics.

The PRISMA 2020 methodology, which involves the studies identification, eligibility criteria establishment, and records screen, is used to select papers [39], as depicted in Fig. 1. More specifically, the indentification of studies is conducted through the databases and two complementary methods. We acquired the related studies on May 21st, 2024, in databases of IEEE Xplore, PubMed, and Scopus, using the advanced search of query keywords in Abstract/Title with ‘EEG’ AND ‘mobile robot/wheelchair/vehicle’ AND ‘control’. Noting that the specific search terms used in each platform were slightly different, and a total of 1731 search results were obtained since 2004. To ensure the review quality that making the most significant research contained, the reports screened had to be published as journal articles. A total of 695 papers were considered and downloaded in Comma Separated Value (csv) format. Duplicate results and non-English articles were then removed. After screening the abstract, the papers focused on the exoskeleton,

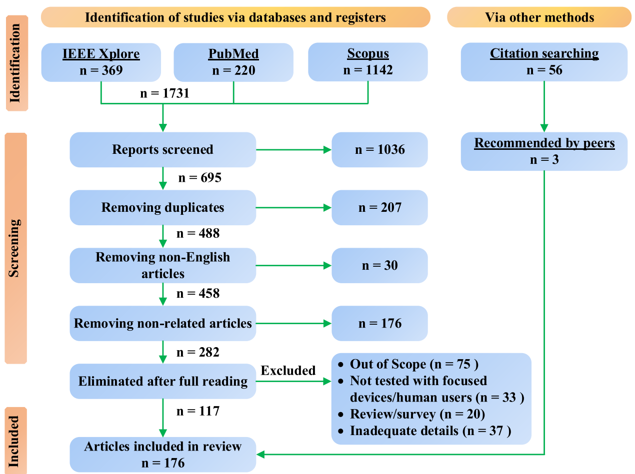

Fig. 1. The paper selection process using PRISMA strategy in this work.

unmanned aerial vehicle, robotic arms, and other not-related fields were deemed irrelevant and excluded. Subsequently, the included outputs were manually refined after the thorough text reading, where the papers were further assessed. Studies of the users' state detection, cognitive load, emotion recognition, attention resource allocation, were excluded as they are out of scope. Specialized review in these regards can be found in [40] and [41]. Further, those review/survey articles, or studies that did not have real-time testing with human users, focused devices, and detailed results were also removed. To ensure that those influential studies, in particular of the highly cited conference papers, are reviewed, 59 papers recommended by peers or from citations were eligible. Following the selection process as shown in Fig. 1, a total of 176 papers were finally included in our current work.

The main contributions of this paper that are not covered by other review papers are three-fold: 1) this paper is the first that comprehensively reviewed on EEG-based intelligent mobile robots from the birth to date, surveying more than 170 relevant studies and classifying representative ones into detailed subcategories, giving a complete and detailed picture in this field; 2) we present a thorough review from the perspective of the targeted systems, key techniques, and performance evaluation, covering the typical and classic exploration at each aspect as much as possible, clarifying the current research status, contributors, and hot topics of the field; 3) the remaining key challenges and possible work are discussed in detail, providing valuable guidance for future research.

C. Paper Organization

This work is structured as shown in Fig. 2. Section II gives an overview of the targeted systems, including the basic concepts of BCI paradigms, typical system types, and relevant research efforts on whole-system design of focused robotics. Section III introduces the status of key techniques concerning the BMIs, control strategies, and robot intelligence. Section IV presents the performance evaluation in the entire process including the participants, experimental protocol, tasks and environments, and evaluation metrics, respectively. Afterwards, the remaining challenges and future research directions are discussed in Section V. The paper ends with conclusions in Section VI.

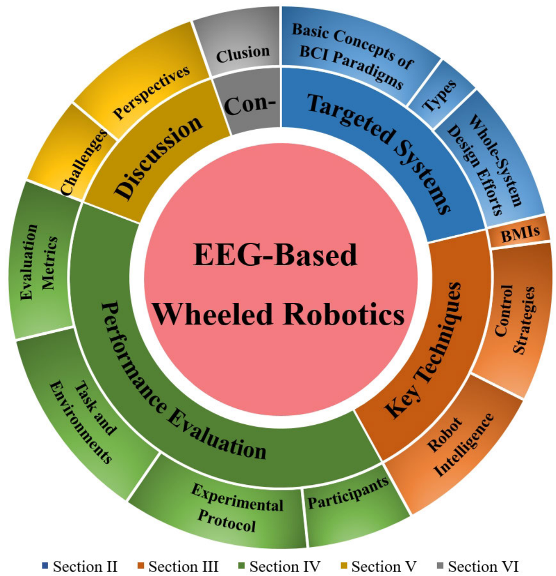

Fig. 2. Structure of major themes focused on the organization of this paper.

II. Targeted Systems of EEG-Based Wheeled Robotics

As a typical hybrid intelligent application, EEG-based robotics have outstanding characteristics of human-machine integration and cover plentiful research aspects of modeling, analyzing, peripheral sensing, control, etc.

As summarized in Fig. 3, the general system architecture of EEG-based wheeled robotics consists of seven basic modules: user interface, user, BCI/BMI, control strategies, environments, robot intelligence, and terminal wheeled robotics. As the thinner arrows of Fig. 3 indicate, to finish the given tasks, the closed-loop system starts with the EEG signals, generated by the user and recorded with the electrodes placed on the scalp surface. Then, the BCI/BMI interprets the specific EEG signals into intentional commands. Three main steps of signal acquisition and preprocessing, feature extraction, and classification are usually identified in a BCI/BMI system. The issued intentional outputs of BCI are further processed, with the help of control strategies, to convert into the control signals (either discrete, or continuous) to actuate the terminal wheeled robotics (e.g., the wheelchair, ground vehicle, mobile robots, and robot swarm). Real-time environments also have an impact on the entire system (see related thicker arrows) and cannot be disregarded when considering the requirements of desired tasks and performance. Thus, robot intelligence, which acts as a medium to sense the environments and intelligently interacts with robots through the control strategies, is involved. Besides, as indicated by left thicker arrows, a user-friendly interface is often integrated to help modulate EEG signals [42], [43], [44], [45], to present real-time environmental information, and feedback the robotic states through a live visual stream (e.g., [36]). Typically, 2-D monitors' screen displays serve as the basis for user interfaces. Bi et al. constructed an effective head-up display (HUD) with a projecting device, windshield, and some attached light reflection films [46]. Based on this interface,

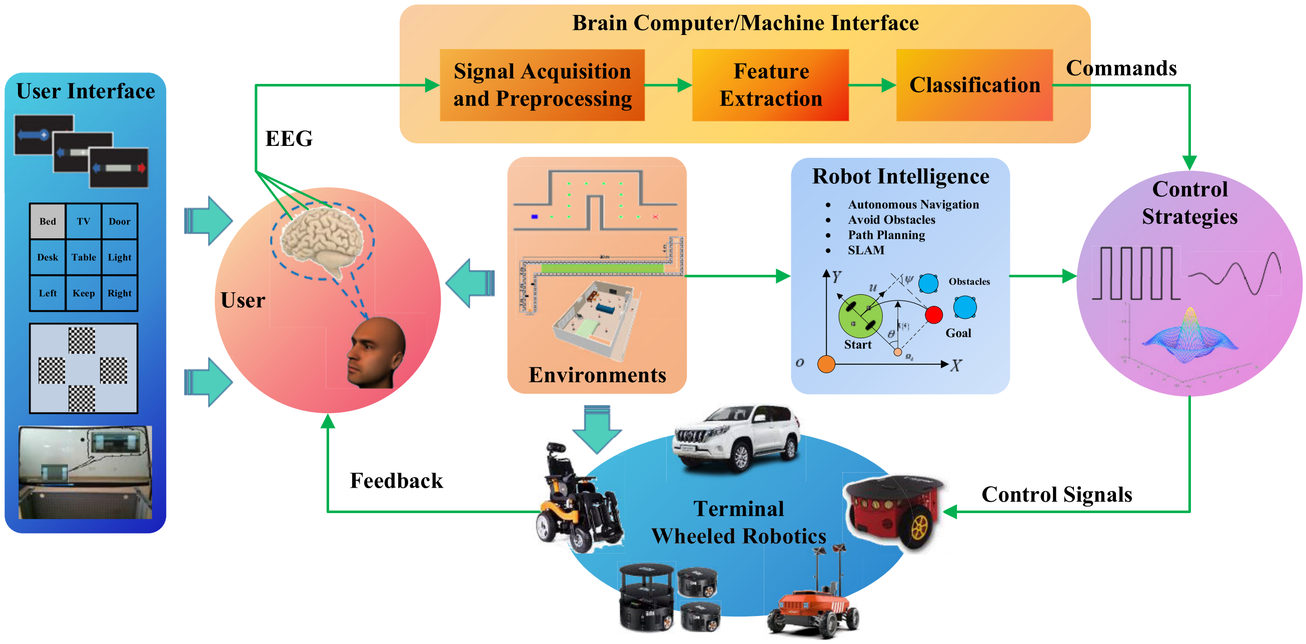

Fig. 3. The general system architecture of seven involved modules for EEG-based wheeled robotics.

many demonstrations of brain-controlled vehicles [19], [47], [48], [49], and EEG-based wheeled robots [50], [51], [52] have been developed. Furthermore, the feasibility of combining BCI/BMI with more advanced technologies has been validated, where studies have constructed user-centric interaction systems with virtual reality (VR) [46], [53], [54], computer vision (CV) [55], [56], [57], [58], and augmented reality (AR) [21], [59], [60], [61], [62], [63].

The BCI/BMI, control strategies, and robot intelligence are the key techniques of the whole system, and we will give a detailed explanation in the next section. Since the module of environments is always related to the task, we will report it in section for performance evaluation. Here, as observed, the modules of the user interface, user, and BCI/BMI are directly coupled to the EEG signals. Thus, in this section, the basic concepts of EEG signals, also known as BCI/BMI paradigms, for EEG-based wheeled robotics are presented to help sort out the general system. Typical system types in terms of the targeted applications, and ongoing research hotspots are also introduced in this section.

A. Basic Concepts of BCI Paradigms

There are three common BCI paradigms: active, reactive, and passive [9]. Active BCI directly and voluntarily decodes a user's conscious brain activity into executable commands. One representative example of signals utilized within this paradigm is event-related desynchronization or ERD and event-related synchronization (ERS), which are elicited by performing mental tasks (e.g., motor imagery (MI), arithmetic operation, mental rotation, word association, or letter composing) [14], [22], [64], [65]. Reactive BCI translates user intention through EEG signals induced with the help of external stimuli, such as visual, auditory, or tactile. Passive BCI monitors the user's mental state via neurofeedback to interact with the external environment, which has been tested for driving [66], [67].

More specifically, the commonly used BCI paradigms within an EEG-based mobile robots are: 1) ERD/ERS signals, such self-paced modulation of brain patterns is characterized by the attenuation or accentuation of the EEG power in specific frequency bands such as μ rhythm (8-13 Hz) and β rhythm (14-30 Hz) activities [27]; 2) P300 Event-Related Potential (P300-ERP) that are elicited in response to infrequent stimuli onset, resulting in a positive reflection at a latency of approximately 300-450 ms [9], [21]. As an example, consider the most typical visual oddball paradigms [43], [44], [57], [68], [69], [70], [71], [72], [73], [74], [75], [76], in which rare and relevant event occurs randomly among the nontarget events. In this paradigm, there is a set of stimuli displayed on a screen and flashing one by one in random order. One of these items is designated as the target on which the user is instructed to concentrate his or her attention (e.g., via a simple way of counting the number of times for flashed target). Then, around 300 ms after the target flashes, a positive potential peak of EEG signals can be determined. In general, the stimuli employed for inducing P300 signals can be visual [21], [42], [43], [44], [57], [63], auditory [77], and tactile [35], [78]; 3) Steady-State Evoked Potential (SSEP) potentials, where signals are steady-state brain responses of evoked potentials by external stimuli at a constant frequency. The utilized stimuli, similar to the P300, can be either visual, which will result in steady-state visual evoked potential (SSVEP) [79], auditory (i.e., auditory steady-state response, ASSR [80]), or vibrotactile (SSSEP in [32]). Especially, SSVEP has been the most favored so far and can be visually evoked with repetitive stimuli of bars, square flickers [81], stripes [5], light-emitting diodes (LEDs) [82], [83], [84], black-and-white checkerboards [47], [85], hybrid [86], etc. 4) Error-Related Potential (ErrP), which is used to indicates the human awareness of erroneous responses, and can be elicited within 500 ms when an error occurred during the interaction with BCI [9], [87]. Since the delayed and non-stationary nature of ErrP slows down the system operation, this paradigm has been most combined within an EEG-based wheeled system in [88], [89], [90], and [91], as an easy-to-use communication channel, to issue yes-or-no commands; 5) Alpha Waves, for which EEG signals at a frequency of 8-12 Hz, may be detected while a person is awake and eyes are closed. Meanwhile, the signals will diminish when the eyes are open or during mental activity. Alpha waves are essentially an idle rhythm signal that is a subset of ERD/ERS; 6) Other EEG signals, such as slow cortical potentials (SCPs), can also be utilized to track users' various intentions [9], [92], [93], [94]. However, because of their high training demand and low ITR, these paradigms have not been extensively used to develop EEG-based robots [25], [26].

In addition to the above single paradigms, the neurological mechanisms of an EEG-based robot may also be hybrid paradigms [9], [95]. A hybrid BCI generally fuses EEG with 1) other sensory inputs, such as electrooculogram (EOG) of eye movements, electromyography (EMG) of muscle activity; 2) other types of brain activity, such as MEG; 3) different BCI paradigms, such as the combination of SSVEP and P300 [20].

B. Typical Brain-Controlled Wheeled Robotic Systems

According to the location of the user and terminal robots, two typical system application types are obtained as the brain-controlled manned systems (i.e., user and robot are co-located), and brain-actuated teleoperated robots (e.g., remotely interaction via telepresence), respectively. Although sharing the same ultimate goal of enhancing mobility, i.e., completing the efficient transmission from the starting point to the destination with brain waves while assuring the overall system performance, there are slight differences in research hotspots between these two systems. Particularly, for manned systems, such as BCWs and BCVs, the safety, user experience, and comfort have more priority, while the brain-teleoperated robots of BCTMs concern more on how the user can effectively operate the robot to finish a given task. Nevertheless, in these brain-robot systems, user intents from the BCI paradigms are often supposed to be one-to-one correlated with the terminal robotic actions, which implies that the robot is expected to execute brain intents accurately in real time. Common activities of these systems include start/stop, forward/backward, left/right turn, acceleration/deceleration, approaching the goal, stay on the given road center, and so on. Several typical systems with the related control functions of mobility and additional intelligence have been listed in Table I, and a more detailed description is given below.

1) Brain-Controlled Manned Wheelchair and Vehicle: In 2005, the first wheelchair operated by a MI-BCI was proposed by Tanaka et al. [16], in which the direction of an electric wheelchair was steered by users via the left and right thinking. Later, many groups joined the research of BCWs. Millán and his group have contributed significantly to the field of EEG-based robots, particularly those based on MI paradigms. Their following research [12], [13], [14], [15], [17], [45], [105], [113], [115], [142], has greatly influenced the development of BCWs in terms of the design of the system architecture, decision fusion, brain-robot interaction, and inspired a large number of breakthroughs in the field of brain-controlled devices [7], [8], [93]. In particular, in addition to the basic robot movement control, they introduced additional intelligence such as path planning, obstacle avoidance, etc., to further ensure the system performance. To decrease the training time while maximizing the accuracy of MI, feedback training methods were proposed by Choi [101]. In 2009, Mandel et al. combined the SSVEP-BCI with an autonomous navigation (AN) system to actuate a real-world wheelchair [82]. Besides, Rebsamen and his colleagues provided the first functioning prototype using P300-BCI in [68], which allows users to freely travel inside the specified office or hospital environments according to the predefined paths. Based on this system, control strategies have been explored in [69] and [70]. For such P300-based BCWs, Minguez's team also devoted themselves to the methodology, users, and experimental verification under unknown and evolving environments [21], [72], [158]. Nunes's laboratory reported great achievements of signal processing [71] and control strategies [44], [74], [119], [120]. Bastos-Filho and coworkers focused on different modalities to command the manned wheelchairs, such as using paradigms of MI-BCI in [159], and SSVEP-BCI in [85] and [126]. For disabled people, they also attempted other different modalities of eye blinks, eye movements, head movements, and sip-and-puff [5].

In addition, several hybrid modalities, such as combinations of MI and SSVEP paradigms [146], and the fusion with different physiological signals such as EEG with cervical movements [149], have been investigated in BCWs. Li's team are highly well-known in this field, with specific paradigms of MI-P300 [141], P300-SSVEP [143], and MI-P300 with eye blinking [145] for EEG-based wheelchairs. Besides, they have also achieved several new brain switch methods [121], shared control strategy [133], and automated navigation techniques [160]. Recent studies are more on the user-centric interaction approach. Yu et al. improved BCWs using a HMD with CV [63], a lightweight robotic arm [56], a novel sequential MI modality [109], and mecanum wheels [151]. Towards the actual use of BCWs, studies have also validated the clinical practicability for patients with severe SCIs [57], [121], [122].

Unlike BCWs, BCVs have greater movement velocities (i.e., about 3-5 m/s [20]) and more complicated dynamics, although the driving speed is still uncomparable to human-driving or automatic-driving vehicles. This makes the related research more challenging while offering a larger range of mobility. The work by Hood et al. in [84], which described the combination of the SSVEP-BCI with a car simulator, appears to be the first study in terms of BCV driving. The preliminary findings provided the feasibility of acceleration and steering via BCI for an on-road car. Göhring et al. pioneered the real automatic driving control through ERD/ERS signals, with two scenarios tested in [18]. Bi et al. tried different BCI modalities of SSVEP with alpha waves [19], P300 [46], hybrid SSVEP-P300 [20], and ERD/ERS with alpha rhythm [147], to select the destination or steer the vehicle directions, respectively. They also led the research on BCV driving experiences in: 1) simulating

TABLE I

REPRESENTATIVE EEG-BASED WHEELED MOBILE ROBOTICS [SIMU: SIMULATED; F: FORWARD; B: BACKWARD; TL/TR: TURN LEFT/RIGHT; ACCE: ACCELERATE; DECE: DECELERATE; DES: DESTINATION; AO: AVOID OBSTACLES; PP: PATH PLANNING; AN: AUTONOMOUS NAVIGATION]

Paradigms	Descriptions	Year	Refs	Systems	Control Functions of Mobility	Additional Intelligence
ERD/ERS	• Endogenous modulation of sensorimotor rhythms; • Limb movement imagining of either left or right, etc. • Allows users to spontaneously deliver the commands at any time by “thinking their thought” voluntarily; • Seems more natural; • Lengthy training time; • Low information transfer rate (ITR) as 0.3 commands per second on average [96];	2004	[11]	Real BCTM	F/TL/TR	AO
		2005	[16]	Real BCW	TL/TR	/
		2007	[12]	Real BCW	F/TL/TR	AO
		2007	[54]	Simu BCW	F/Stop	/
		2009	[53]	Simu BCTM	TL/TR/Acce/Keep	/
		2010	[15]	Real BCTM	F/TL/TR	AO
		2010	[97]	Real BCTM	F/TL/TR/Stop	/
		2011	[98]	Simu BCTM	F/TL/TR/B	Stop
		2011	[99]	Real BCW	Three commands	SLAM
		2011	[100]	Simu BCW	TL/TR	AO, Wall following
		2012	[101]	Real BCW	F/TL/TR	/
		2013	[17]	Real BCW	F/TL/TR	PP, AO
		2013	[18]	Real BCV	TL/TR/Speed/Slowdown/Keep	AO, Path tracking, Stop
		2013	[102]	Real BCW	F/TL/TR/Left/Right diagonal	/
		2013	[103]	Simu/Real BCW	F/TL/TR/B	/
		2014	[59]	Real BCTM	TL/TR/Stop	AO
		2014	[104]	Simu/Real BCTM	F/TL/TR/B/halt/Main	PP, AO
		2015	[105]	BCTM	TL/TR/Keep	AO
		2016	[106]	Simu BCTM	F/TL/TR	AN, AO, Wall following
		2016	[107]	Real BCW	F/TL/TR/B/On/Off	/
		2018	[108]	Real BCW	F/TL/TR/B/Stop	/
		2018	[109]	Real BCW	F/TL/TR/Acce/Dece/Stop	/
		2018	[110]	Real BCTM	F/TR	SLAM
		2019	[111]	Real BCTM	TL/TR (anti/clockwise)	SLAM, PP
		2019	[112]	Simu BCTM	TL/TR/Acce/Dece	/
		2020	[113]	Real BCTM	TL/TR	/
		2021	[114]	Real BCTM	TL/TR/Acce/Stop	AO
		2022	[115]	Real BCTM	TL/TR	AO
		2023	[116]	Real BCW	F/TL/TR/Stop	AO
P300-ERP	• Evoked; • Relatively evident signal patterns; • Do not require extensive training; • A few minutes is enough to calibrate the parameters of the detection algorithm; • High accuracy of above 90%; • Long command issuing interval causing a low ITR;	2006	[68]	Real BCW	Des Selection	AN
		2007	[117]	Real BCW	Des Selection	AN, Stop
		2008	[71]	Real BCW	Eight motion directions, Stop	/
		2009	[118]	Real BCTM	Desired motion direction, Stop	/
		2009	[72]	Real BCW	Nine Des Selection	AN, AO
		2011	[119]	Real BCW	F/TL/TR/Stop/B/3 Des/4 other	PP, AN, AO
		2012	[21]	Real BCTM	Nine Des Selection	PP, AN, AO
		2013	[46]	Simu BCV	Nine Des Selection	/
		2013	[120]	Real BCW	F/TL/TR/B/Stop	PP, AN, AO
		2014	[35]	Simu BCW	F/TL/TR/B	Stop
		2016	[74]	Real BCW	F/B/TL/TR/Stop/WC/Exit	PP, AO, SLAM
		2017	[121]	BCW	Des Selection/Start/Stop	PP, AN, Stop
		2017	[122]	Real BCW	Des Selection	AO
		2017	[123]	Real BCTM	F/TL/TR/B	AN
		2017	[76]	Real BCTM	F/TL/TR/Stop	Stop
		2018	[56]	Real BCW with Arm	F/TL/TR/Left/Right translation	PP, AN
		2020	[43]	Real BCTM	F/TL/TR/B	Stop
		2021	[44]	Real BCW	F/B/Stop/Left/Right 90/WC/HELP	Motion planner, SLAM
		2022	[63]	Real BCW	F/TL/TR/B/Left/Right translation	AN, AO
		2022	[57]	Real BCW	F/TL/TR/Left/Right translation	/
SSVEP	• Evoked; • Produced by blinking stimulus visualization; • Relatively high accuracy of above 90%; • Less or even no training; • Adequate commands	2008	[124]	Simu BCW	Nine motion direction, Stop	/
		2009	[82]	Real BCW	F/TL/TR/B	AN, AO
		2010	[83]	Real BCTM	F/TL/TR/B/Stop	/
		2010	[85]	Real BCW	4 Destinations	PP, AN, AO, Path tracking
		2012	[125]	Real BCTM with Arm	F/TL/TR/B/move robotic arm	/
		2012	[84]	Simu BCV	F/TL/TR	/

TABLE I

(Continued.) REPRESENTATIVE EEG-BASED WHEELED MOBILE ROBOTICS [SIMU: SIMULATED; F: FORWARD; B: BACKWARD; TL/TR: TURN LEFT/RIGHT; ACCE: ACCELERATE; DECE: DECELERATE; DES: DESTINATION; AO: AVOID OBSTACLES; PP: PATH PLANNING; AN: AUTONOMOUS NAVIGATION]

	number; • High ITR; • May provoke fatigue after long-term use;	2013	[126]	Real BCW	F/TR/TL/Stop	/
		2014	[5]	Real BCW	F/TR/TL/Stop	PP, AN SLAM
		2015	[127]	Real BCW	F/TR/TL/Stop	Wall following, AO
		2015	[128]	Real BCTM	d /θ increase/decrease	/
		2016	[129]	Real BCTM	F/TL/TR/Change mode	/
		2017	[36]	Multi BCTM	F/TL/TR/B	PP
		2017	[130]	Real BCW	TL/TR	PP, SLAM, AO
		2018	[131]	Real BCTM	F/TL/TR/Stop	PP, SLAM, AO
		2018	[81]	Real BCW with Arm	F/TL/TR/B	/
		2019	[48]	Simu BCV	TL/TR/Acce/Dece/lane keeping	/
		2019	[132]	Real BCTM	F/TL/TR/Stop	AO, PP
		2020	[49]	Simu BCV	TL/TR	AO
		2020	[133]	Real BCW	F/TL/TR/Dece	PP
		2020	[60]	Real BCTM	F/TL/TR	Augmented Reality
		2021	[37]	Multi Real BCTMs	Des Selection	AO
		2021	[38]	Multi Simu BCTMs	TL/TR/Keep/Acce/Dece/Stop	/
		2022	[52]	Simu & Real BCTM	F/TL/TR	AO
		2024	[134]	Simu BCTM	F/TL/TR/Acce/Dece/Stop	AO
SSSEP	• Induced at one's own will by tactile stimulation	2015	[135]	Real BCW	F/TL/TR	/
		2016	[136]	Real BCW	F/TL/TR	/
		2018	[33]	Real BCW	F/TL/TR	/
Errp	• Delayed and non-stationary nature;	2019	[89]	Simu BCW	Yes/No	PP, AN, AO
		2021	[91]	Simu BCW	Yes/No	PP, AO
Alpha	• By instructing users to close and open their eyes;	2009	[137]	Simu BCTM	F/TL/TR	/
		2010	[138]	Simu BCTM	TL/TR/Stop	AN
		2016	[139]	Real BCTM	F/TL/TR	/
Hybrid	MI-P300	2010	[70]	Real BCW	Des Selection, Stop	PP, Stop
	SSVEP-ERD/ERS	2011	[140]	Real BCW	F/TL/TR/Stop	/
	MI-P300	2012	[141]	Simu/Real BCW	TL/TR/Acce/Dece	/
	MI-EMG	2013	[142]	Real BCTM	TL/TR/Keep/Stop/Start	AO
	P300-SSVEP	2013	[143]	Real BCW	Go/Stop	/
	MI-EMG	2013	[144]	Real BCW	F/TL/TR/Stop	/
	SSVEP-α waves	2014	[19]	Simu BCV	F/TL/TR/Start/Stop	/
	MI-P300-EOG	2014	[145]	Real BCW	F/TL/TR/Acce/Dece/B/Stop	/
	MI-SSVEP	2014	[146]	Real BCW	F/TL/TR/Acce/Dece/Start/Stop	/
	MI-α waves	2014	[147]	Simu BCV	TL/TR/Start/Stop	/
	P300-SSVEP	2015	[20]	Real BCV	Nine Des Selection	/
	ERP-EOG	2015	[148]	Real BCTM	Direction/F/Stop	/
	SSVEP-cervical move	2016	[149]	Real BCW with Arm	Mode Selection/F/TL/TR/B/Grasp	/
	EEG-EMG	2016	[150]	Real BCW	F/TL/TR/B	Stop
	MI-P300	2017	[151]	Real BCW	11 commands	/
	MI-SSVEP	2017	[152]	Real BCW	At least 8 commands	/
	MI-EMG	2017	[153]	Real BCW	F/TL/TR/Stop	Stop
	MI-SSVEP	2018	[58]	Simu BCV	F/TL/TR/Start/Park	Computer vision-based AN
	VEP-Eye tracking	2019	[154]	Real BCW	F/TL/TR/B	/
	EEG-EOG	2019	[155]	Real BCTM	F/TL/TR	SLAM, Stop
	MI-Head move	2020	[156]	Real BCV	TL/TR/Acce	/
	MI-EMG	2022	[157]	Real BCTM	F/TL/TR	/

and predicting the driving performance [161], [162], [163]; 2) designing driver-vehicle interface [164]; 3) developing control strategies to improve the performance within the control of lateral [49], longitudinal [47], and both [48]. In 2018, Li et al. developed a hybrid SSVEP-MI BCI to interpret the user's intention for a simulated BCV, where a CV based automatic driving component was combined to process the complex road situations [58]. To alleviate manual activities in fieldwork, an interesting EEG-based tractor-driving robot with assistant control was developed by Lu et al. [165].

2) Brain-Actuated Teleoperated Mobile Robots: Compared to the manned BCWs and BCVs, BCTMs offer end-users remote mental control capacity with wheeled robots, and has been flourished for nearly two decades. Early tele-systems adhered to the idea of the first EEG-based robot [11], focusing on the basic motion control with ERD/ERS [53]. Barbosa applied four mental imaginary activities of tongue, feet, left and right arms to command the mobile robot's forward, left and right turn, and stop movement [97]. The proposed method yielded a high rate of successful commands. In [166], a typical rehabilitation robot was built by continually monitoring users' intentions with ERD/ERS, and the proposed method was verified to help ensure patients' active participation. As the pioneer in the field, Millán et al. also developed a MI-based telepresence robot with a webcam and laptop to facilitate the remote brain-robot interaction [105]. The system has been investigated with a shared control approach [10], [15], hybrid BCI modality [142], and tested with disabled users [105], to assist in completing a rather complex task. Moreover, they have proposed a variety of middleware to further enhance the human-robot interaction, such as with AR [59], robot operating system (ROS)-neuro packages [167], [168], continuous control framework [113], and shared intelligence systems [115].

The feasibility of a P300-based telepresence robot, for patients with severe ALS disabilities, has been reported by Minguez's team [73]. With access to the Internet, the same system has been further tested in remote environment [21]. Li's group has been working on the teleoperation of SSVEP-based mobile robots. They have conducted a number of research on the incorporation of techniques such as the Bezier curve [128], deep learning (DL) [131], simultaneous localization and mapping (SLAM) [110], [130], [132], and artificial potential field (APF) [130], [131], [132], [169], [170]. In the same group, the technology of DL-SLAM for MI-based robot teleoperation [111], and the viability of brain multi-coordinated robotic systems with visual compressive feedback [36] have been investigated. Besides, Si-Mohammed et al. have successfully verified the feasibility of the AR technique for SSVEP-based remote robots to provide a first-person control perspective [60]. There is another group led by Bi focusing on SSVEP-actuated wheeled robots. To enhance the system performance, different control strategies including model predictive control (MPC) [50], sliding mode control (SMC) [51], [52], and fuzzy logic control (FLC) [171], have been investigated. They also tried SSVEP based lead-follower formation control of multiple robots [38]. Dai et al. have also presented a layered shared framework for EEG-based brain-multirobot cooperation [37]. The findings showed the control approach enables the robot swarm to complete all tasks effectively, safely, and flexibly.

C. Main Research Efforts on Whole-System Design

As mentioned above, the great majority of research of this field have used existing EEG amplifiers, computers/laptops, robots, and sensors to construct the system, and then develop relevant BMIs and control strategies [172]. Nonetheless, from the standpoint of the fused system as a whole, there are also quite a lot of researchers that conducting studies on robots' mechanical structure design and modification, and model simulation, to facilitate the brain-robot symbiosis.

1) Mechanical Design and Modification: An off-the-shelf device of commercial electric wheelchairs [17], [69], [72], robots [36], [52], [105], or vehicles [93], [165], is generally served as the basic system component. Meanwhile, a number of sensors are often integrated for perceiving peripheral and self-state information. For conventional BCWs, position adjustments in moving direction are along the wheel direction, causing slight adjustment in other directions time-consuming and even exhausting. An omnidirectional chassis with Mecanum wheels has thus been developed to enable good mobility in restricted narrow space [56], [57], [109], [151]. Besides, Utama et al. tested the load capacity of a wheelchair for BCI users. Their involvement in the variation relationship among velocities, response time, and loads opened up new avenues for BCWs [173]. For instance, how to avoid rapid velocity fluctuations in response to varied BCI commands, given that accuracy is not exactly 100% [114]. Moreover, to enhance the system capability of going somewhere and performing some tasks, some retrofitted wheelchairs with mounted robotic arms have been developed for BCTMs [174], and BCWs [56], [57], [63], [81], [149].

2) Model Simulation: Modeling and simulating the behaviors of an EEG-based wheeled robot has been investigated. An interesting exploration of a computer simulation of artificial SSVEP has been conducted by Lee et al. [175]. Bi et al. conducted a pilot study for steering control of a SSVEP-BCV using an extended queuing network which incorporated a brain-controlled decision model, a BCI performance model, and a vehicle and interface model [162]. Experimental results showed that the performance of simulated closed-loop brain steering behaviors was close to that of real drivers. To better understand the brain-control behaviors of the simulated vehicle, a mathematical encoding model [163] and a human behavior model of longitudinal driving [161] were developed. In the same spirit of queuing network modeling, Li et al. is likely to be the first to investigate how to decouple the user from the closed loop of brain-controlled mobile robot, where a pair of simulated operator decision model and BMI performance model were proposed to mimic the user's steering control [176].

III. Key Techniques of EEG-Based Wheeled Robotics

This section will give detailed review on the main progress for each research area of EEG-based wheeled robotics.

A. Brain-Machine/Computer Interfaces

The core of a BMI/BCI for EEG-based robots is to interpret user neural signals into the tangible commands that can be executed. As a result, the key techniques of a BCI lie in how to acquire and process the neurological brain activities.

1) Signal Acquisition: The acquisition of EEG signals requires the use of specialized devices, which consist of several electrodes placed on the surface of scalp. Although electrodes of copper [146], or gold [103], were used in a few studies for EEG raw signal recording, currently silver/silver chloride (Ag/AgCl)-based nonpolarized electrodes have been the most commonly utilized because of their relatively low contact impedance, better stability, and low cost [71]. These electrodes are often deployed in accordance with a standard, such as the 10-10 system [152], [177], [178], or the most frequently used

10-20 international system [42], [142], [149], [179], depending on where the actual distance between adjacent electrodes (10% or 20% of the total distances of reference sites) [22].

To increase the conductivity, conventional electrodes require the removal of adhering epidermal keratin before use and the insertion of a conductive liquid such as gel and saltwater between them and the scalp, thus being termed as the "wet" electrodes. Since the above operations may take a longer time and are more likely to cause discomfort to the user, several "dry" electrodes without skin cleaning and conductive gels have been developed [180], [181]. Though most current EEG-based robots use wet electrodes, dry EEG electrodes based on novel materials are rapidly advancing the BCI fields [86], [182], [183], [184], achieving comparable signal-to-noise ratio (SNR) and classification accuracy in stationary environments [185]. In terms of practical application, except for pasting each electrode manually, a variety of commercial items (e.g., NuAmps from Neuroscan Inc., g. USBamp produced by g. tec) has already successfully been applied within the fields. Detailed recording devices can be referred from Table II, which also lists the most used common algorithms of the following signal preprocessing, feature extraction, feature selection, and classification.

2) Signal Preprocessing: The preprocessing step is always required to improve the quality and SNR of acquired raw signals, which generally accompanied by the power line noise, blink/movement-related artifacts, etc. As shown in Table II, the efficient and most commonly used methods is the frequency domain filtering, where the highpass, lowpass, bandpass (e.g., 0.1∼100 Hz), and notch filters (generally at 50 and 60 Hz) are employed to remove the line and specific frequency noise. As a result, most of unphysiological noise and several physiological artifacts such as high-frequency EMG (almost over 100 Hz), and low-frequency electrocardiography (ECG) signals can be effectively attenuated. To improve local response of a specific cortical region, spatial filtering techniques, such as Laplacian filtering and common average reference (CAR), for which the average value of all channels is subtracted from one interested channel, have also been preferred. The canonical correlation analysis (CCA) measures the underlying correlation between two sets of multi-dimensional variables, and the minimum energy (ME) approach, independent component analysis (ICA), principal component analysis (PCA), and common spatial patterns (CSP) have also been utilized in EEG-based wheeled robots. Other methods being applied to attenuate the artifacts include the temporal filter, blind source separation (BSS) algorithm, wavelet analysis. Details of these methods used for EEG preprocessing can be found in [207] and [208].

The strategy of frequency domain filtering, however, has an apparent disadvantage of filtering out important EEG signals that are on the same frequency as the artifacts. Meanwhile, the EOG artifacts which often has large amplitude, randomness, and generally overlap with the frequency of useful EEG. All these causing it difficult to perfectly rejecting unexpected noise. Therefore, in the real-time EEG-based robotic systems, users are often requested to minimizing the movement of their body and eyes. In addition, each method mentioned above has its pros and cons, for example, the Laplacian derivation is simple and easy to use while not very robust. By contrast, CSP is more robust but requires a larger number of channels. Compared with simple filtering, the methods of ICA and PCA have higher algorithmic complexity and are more difficult to use. Therefore, the preprocessing methods need to be designed properly, especially for those brain-controlled systems which has the strict requirement of the command issuing interval. To this end, a method combining wavelet and CCA has been proposed to effectively achieve removal of EOG artifacts while preserving EEG components [114]. In [203], the influence of the ocular and muscular artifacts was explicitly considered, with relevant online removal methods of based on ICA and adaptive noise cancellation was applied for practical ERD/ERS-based vehicle.

3) Feature Extraction: To obtain a satisfactory classification performance, distinguishable features of frequency, temporal, and spatial domains are often extracted in signal processing. In general, when the user performs a task that modulates the power spectrum component of EEG signals, the features in the frequency domain are calculated. In real EEG-based robotics, the power spectral density (PSD) has been most estimated as features using Fourier transform (FT) of auto-correlation function (ACF) [36], and Welch's periodogram. Thanks to the digital technology, the vast majority of FT methods employ discrete Fourier transform (DFT) [82], [143], or more precisely, fast Fourier transform (FFT), which is a fast and effective computing method in practice to execute the DFT. Wavelet transform (WT) is an adjustable-window Fourier analysis to provide flexible time-frequency resolution. Compared with the FFT, Hilbert-Huang transform (HHT) is self-adaptive and more suitable to tackle nonstationary EEG signals with a better result in real-time robotics [65]. Two main processes of the empirical mode decomposition (EMD) and Hilbert transform are often needed within the HHT, and to enable the small SSVEP-based robot, an ensemble EMD, which iteratively performs the sifting procedure and takes the ensemble average as the final output, has been developed in [175]. In addition to PSD, the value of logarithmic band power (BP) has also been used to estimate the EEG power. Since only squaring, averaging, and logarithmic operations are involved, this method is well suited for the real-time applications.

Furthermore, the common spatial patterns (CSP) method has also been frequently used, for which the basic principle is to find a set of optimal spatial filters for projection so that the difference between the variance values of the two types of signals is maximized, thus obtaining a feature vector with a high degree of discrimination. There are numerous successful applications within EEG-based robots of feature extraction by CSP and its improved methods, such as multi-CSP, common spatial frequency subspace decomposition (CSFSD). Another spatial filtering approach CCA can be utilized to extract EEG features [133], [202]. If the brain response is evoked in the temporal domain as P300 paradigm, the statistical features of the signals across time (values of amplitude, mean, median, variance, latency and topographic distribution, etc.) are usually extracted by signal averaging and moving averaging techniques. Besides, other features

TABLE II
MOST COMMON DEVICES/METHODS EMPLOYED WITHIN EEG-BASED WHEELED ROBOTICS

Steps	Category	Description	Examples
Signal Acquisition	g.tec	Austria	g. USBamp [21], [35], [42], [44], [71]-[73], [82], [83], [102], [104], [105], [113], [115], [119], [120], [129], [148], [167], [174], [186], g. BSamp [98], [103], [159], g. GAMMA cap [89], [187], g. Nautilus [43], g. Sahara [152], g. GmgH [188], and amplifiers [15], [54], [100], [101], [137], [138], [146], [177], [178], [189]-[191]
	Neuroscan	Australia	Bioamp [175], NuAmps [68]-[70], [110], [111], [117], [121], [128], [131]-[133], [141], [143], [145], [160], [169], [170], [192], Siesta [65], SynAmp2 [81]
	Brain Product	Germany	BrainAmp [37], [135], actiCHamp [56], [57], [63], [116], [179], [193]
	SYMTOP	Chinese	[19], [20], [46]-[52], [147]
	Emotiv	USA	EPOC caps [18], [107], [114], [123], [149], [153], [154], [157], [165], [194]-[196], INSIGHT [197]
	Neurosky	USA	[195], [196], Mind Wave [198]
	Biosemi	Netherlands	[13], [14], [91], [99], [140]
	EMSA Equip	Brazil	BrainNet-36 [85], [140]
Signal Preprocessing	Frequency domain	Highpass filter	Filtered at 1 Hz [14], f_p = 200 Hz [157]
		Lowpass filter	35 Hz [97], 40 Hz [152], 50 Hz [149], and [140], [199]
		Bandpass filter	0.1~100 Hz [15], [128], [140], 0.1~30 Hz [71], [148], 0.5~30 Hz [20], [21], [72], [159], 0.5~60 Hz [83], 3~20 Hz [125], 8~30 Hz [101], [191], 15~100 Hz [138], and [200], [201]
		Notch filter	48~52 Hz [76], 50 Hz [15], [50], [58], [65], [71], [101], [120], [146], [167], [171], [191], 60 Hz [33], [135], [136]
	Spatial domain	Laplacian filtering	[15], [17], [83], [105], [113], [115], [142], [167], [188]
		CAR	[14], [84], [85], [112], [144], [160], [192], [201]
		Others	CCA [58], [114], ME [82], [83], [129], [143], [202], ICA [203], PCA [68]
	Other	-	Temporal domain [5], CSP [100], wavelet analysis [114], [165], BSS [101], [191]
Feature Extraction	ERD/ERS	PSD	[14], [105], [116], [179], [188], [204]
		FT	FFT [16], [70], [107], [179], WT [53], [97]
		Welch's periodogram	[11], [14], [15], [105], [113], [115], [142], [159], [167], [188]
		Logarithmic BP	[13], [54], [177], [178], [187], [189], [190]
		CSP	[53], [58], [100], [101], [106], [110]-[112], [114], [141], [144]-[146], [160], [187], [191], multi-CSP [109], [151], CSFSD [101]
		Others	HHT [65], AR [205], AAR [159], Hjorth parameters [205]
	P300	-	PCA [20], WT [42], Signal average [141], [143], Moving average [21], [72], [73]
	SSVEP	FT	FFT [47], [48], [83], [128], [149], DFT [82], [143]
		Welch's periodogram	[127], [131]
		Others	CSP [47], EMD [175], CCA [133]
	Other paradigm	-	FFT [33], CSP [33]
Feature Selection	-	CVA	[13], [17], [45], [105], [113], [142]
		Others	PCA [20], [46], Neighborhood component analysis [43]
Classification	ERD/ERS	k-Nearest Neighbor	[58], [205]
		LDA	[13], [98], [103], [109], [141], [147], [151], [177]-[179], [187], [189], [190]
		SVM	[101], [106], [110]-[112], [114], [116], [144], [146], [153], [159], [160], [191]
		Statistical Gaussian	[11]-[15], [17], [105], [113], [115], [142], [167]
		ANN	[65], [97], [100], [107], [108], [153], [157]
		Others	Simple linear classifier [54], Recursive training [16], MLD [188]
	P300	LDA	[20], [42], [44], [46], [148], BLDA [91], [122], SWLDA [21], [56], [57], [63], [72], [73], [200]
		SVM	[43], [68]-[70], [117], [121], [141], [143], [145], [160]
		Statistical Gaussian	[71], [119], [120]
	SSVEP	LDA	[20], [83], [202]
		SVM	[19], [36], [47]-[52], [81], [128], [131], [171]
		CCA	[38], [58], [81], [86], [127], [130], [132], [133], [146], [174], [193], [192], [201], [206]
		MSI	[37], [132], [169], [170]
	Other paradigm	LDA	[19], [147], RLDA [33], SWLDA [35]

such as autoregressive (AR), adaptive autoregressive (AAR) parameters, and custom algorithms, such as the tuned-residue iteration decomposition (t-RIDE) [43], [76], have also been attempted in feature extraction.

4) Feature Selection: Due to the considerable amount of information included in EEG, the above extraction procedure may result in a large set of feature vector. Besides, to improve the subsequent classification accuracy, a combination of some features may be performed. Excessive combinations, however, also lead to redundancy, overfitting, and increased processing time. Thus, before the classification stage, a further procedure of feature selection may be required to remove irrelevant features while keeping the most discriminable ones. Based on cross-validation of training data, canonical variate analysis (CVA) is often used to select the user-specific features that maximize the differentiation across tasks. The PCA is preferred in dimensionality reduction to ensure real-time performance.

5) Classification: Various classification approaches such as the linear classifiers (e.g., simple threshold [54], [73], [76], [82], regularized linear logistic [89]), and nonlinear methods have been used to real-time EEG-based systems. Nearest neighbor classifiers are very simple in that they assign a feature vector to a class based on its nearest neighbor. With the majority of the nearest k neighbors, k nearest neighbor (kNN) has been the most frequently used. Linear discriminant analysis (LDA), also known as Fisher's LDA, is another simple-to-use classifier, where the philosophy can be summarized as distinguishing classes with the minimum intra-class variance and the maximum inter-class variance. This classifier has a relatively low computing requirement, making it appropriate for a wide range of online BCI systems. The main drawback of LDA is its linearity, which may yield poor results on large nonlinear EEG data. To solve this problem, regularized linear discriminant analysis (RLDA) was developed based on a kernel function of modified samples covariance matrix. The extended LDA of Bayesian LDA (BLDA) and step-wise LDA (SWLDA) have also been extensively employed to improve the classification performance. In [152], a quadratic discriminant analysis (QDA) is used, which is a generalized LDA and the assumption that the same covariance for each class in not considered. Furthermore, support vector machine (SVM) identifies different classes using a discriminant hyperplane that is selected to maximize the margin (i.e., the distance between the nearest training points) and bears the advantages of a few hyperparameters defined by hand, insensitive to overtraining, and good generalization properties. These benefits, however, come at the tradeoff of a low execution speed than others.

Another largely used category of statistical classifiers usually select the highest probability from an estimation of the posterior class probability distribution and are able to represent the EEG uncertainty. Artificial neural networks (ANN) are able to simulate human cognition process with several artificial neurons, and are successfully applied in ERD/ERS with good performance. For the SSVEP, two commonly utilized methods are CCA, and multivariate synchronization index (MSI). Typical applications of above classifiers used in the field can be referred in Table II. Besides, decision trees [5], [85], [153], fuzzy logic [107], Mahalanobis linear distance (MLD) [188], and Hidden Markov Model (HMM) [209] have also been used.

Bearing the advantages of powerful feature learning capacity, the advanced deep learning algorithms, such as convolutional neural networks (CNNs) [114], [116], [184], [194], [199], and recurrent neural network (RNN) [165], have also been adopted recently. With the help of transfer learning, the time cost of the used algorithm in [165] is 0.48 ms, showing great potential in real-time application of EEG-based robotics. To achieve better performance to suit the online purpose, optimization algorithms have been considered, where the thresholds of the developed classifier were chosen for each category in [189], and the window length was optimized using offline training data [109]. In [100], the particle swarm optimization (PSO)-R^2 method is applied for fast spatio-spectral and temporal parameter optimization of the ANN classifier, while a fuzzy PSO with cross-mutated operation was considered for ANN in [65]. More details of the parameter optimization for the commonly used classification methods above can be seen in reviews of [208].

Finally, to verify the effectiveness of the classifiers, the statistical measures, such as accuracy, true positive ratio (TPR), and false positive ratio (FPR) are generally calculated [145], [170]. The correct recognition is represented by a positive and thus, TPR indicates how many positives the classifier is able to detect. Instead, the misclassifying from false features into true results may lead to a false positive and FPR means the rate of those detections. Indeed, there are also two related metrics of true negative and false negative, which can be referred in [28]. Once the outcome of the classifiers satisfies the requirement, it can be used for the subsequent online experiments.

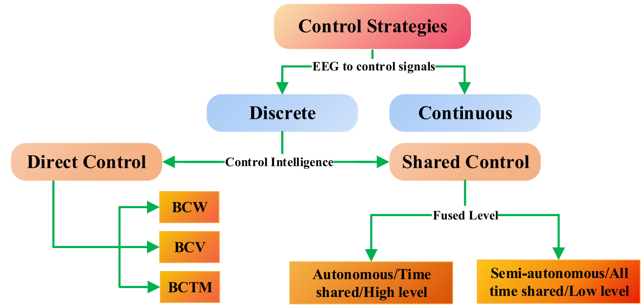

Fig. 4. Tree diagram depicting the categorization of control strategies.

B. Control Strategies

To bridge the brain and robot system, firstly, the relationship between elicited mechanisms and instruction set is often predefined. Then, from the induced EEG, different control strategies are applied to realize the multi-objective robot control of start and stop, direction, speed, etc. The control techniques may be divided into several categories according to the practical driving principle, or control signal producing mechanism, as shown in Fig. 4.

1) Discrete Control Strategy: After the BCI/BMI module, a straightforward and most universally employed control strategy, which converts the user intention to tangible control signals, is directly mapping the classification commands to the control commands. For this control strategy of command-to-movement scheme, the sparse commands issued by BCI will lead to a nature discrete interaction modality between the user and robots [29]. The resulting commands allow direct robot operation, which is known as the operational mode of direct control by BCI [22]. For the EEG-based robots system under this control strategies, the users are engaged in the whole brain-controlled process with determining the each robot movements. Tanaka et al. employed this control strategy to demonstrate the direction control of the first BCW, where the users were asked to think the left or right direction to control the wheelchair turn left or right [16]. Leeb and coworkers investigated the go and stop control of a BCW in virtual environment [54]. Several hybrid BCI-based systems have also applied direct control strategy [19], [141], [146], [152].

The strategy of direct control by BCI does not require extra control intelligence, which simplifies the system's complexity while also endowing the user a full brain control authority. However, the task execution effects of such system are completely dependent on the BCI performance [22]. Moreover, during the brain-controlled process, the users have to frequently deliver the commands while simultaneously keeping in working memory the execution of the desired mental task and the details of the environment that may require issuing another command. Even in a basic point-to-point motion activity, the user needs to be always alert for sudden collisions with unexpected obstacles, which may cause rapid fatigue and prevents long-term operation.

Therefore, some control intelligence, mainly related to the robot, was incorporated to alleviate the users' cognitive workload during interaction and enhance the system performance. This is known as the strategy of shared control [11], which integrates both the human and machine intelligence, and thus allowing a better control of the bio-robot. The significant advantages of the shared control strategy are as follows. First, machine intelligence can detect environmental events and automatically avoid danger under emergency situations, ensuring the system safety, which is particularly critical for the manned BCWs and BCVs [14]. Second, with the assistance of machine intelligence in coping with low level navigation issues, the control accuracy is enhanced and the system is hence able to work in more complex tasks and surroundings [17]. In addition, as compared with the direct control by BCI, the use of shared control allows the user to finish a given task with fewer commands, thus improving the execution efficiency and extending the using time of the EEG-based robotic system [15], [55].

According to the effective mechanism and fused level, the shared control strategy may be further categorized into autonomous and semi-autonomous [15], [70], or time-shared control and all-time shared [133], or high-level and low-level [44]. The autonomous/time shared/high level ones refer to the strategy that the user and robots work at different periods. More specifically, the user just needs to indicate the final destination or a staged direction to interact with the robot, whereas the subsequent executing processes (e.g., autonomous navigation or a fixed displacement) are finished by the robot. Rebsamen et al. proposed a representative example using this strategy for BCWs [68], [69], [70]. In their system, the user input was limited to destination selection via P300 signals from a set of possible rooms (e.g., lab, office, toilets, lift, etc.), which were icons displayed on a graphical user interface. Then, the BCW movement will be performed automatically until the guiding paths preprogrammed with a path editor finished. Minguez's team developed an exploration mode where a model builder based on camera was designed to construct the real time local environmental model with both static and dynamic objects [21], [72], [73]. Then, the possible destinations presented by a polar grid in graphical interface were able to be selected for the P300 user. Other examples for the control strategy of such a destination selection approach can be referred in BCWs [42], [44], [57], [89], [121], [122], [160], BCVs [20], [46], and BCTMs [128], [199]. In a nutshell, this type of autonomous EEG-based systems greatly reduces the operational stress, with the extremely control authority sacrifice since users are totally unable to intervene during the execution procedures.

As the leader of the system, users, however, always wish to keep as much control as feasible. Thus, a semi-autonomous/all time shared/low level control strategy seems generally more suitable and ecological. For such a shared framework, the user and control intelligence simultaneously collaborate to operate the robot. One of the modalities for the tradeoff between these two agents is that the user is involved in the motion control of the EEG-based robots most of the time, while the control intelligence only assists users to deal with problematic situations, such as obstacle avoidance. A basic but effective control intelligence is to supplement the direct BCI control with always sensing the robot's surroundings and immediately stop the system when an obstacle detected in front of it [11], [35], [55], [76], [142], [153], [173], [198]. However, frequent stopping actions of above strategies not only require additional assistance to reposition the robot, but may also frustrate the user. Several other shared control methods hence being inspired, when a collision may occur, to assist the user by making the robot automatically turns towards the opposite direction [15], or moving along the edge of obstacles [179], or adjusting commands with designed rules [114], [194]. Millán and coworkers developed an assistive algorithm to determine the output commands at each sparse time instant. They estimated the user's steering intents with the probability distribution that generated with the environmental information. Then, the possible wrong commands from BCI will be filtered out with this contextual control strategy. According to the experimental results [12], [13], [14], the overall driving performance has been significantly improved.

Different with the above simple replacement, the outputs of user and robot intelligence can also be fused with linear weights formulation [37], [130], Bayesian approach [133], [210], logical relation [58], and a finite state machine (FSM) [127], [200]. Moreover, there are a large amount of more advanced control intelligence that has been widely explored. Model predictive control (MPC) is a promising approach, which incorporates multiple control objectives and constraints into an optimization problem. Considering the constraints of vehicle dynamics and road safe region, Bi et al. proposed assistive MPC controllers for simulated BCVs. The lateral steering control [49], and the combined lateral and longitudinal control [48], have demonstrated the viability of such collaborative mechanism for BCVs. To ensure the safety of EEG-based mobile robot in an open space, they have integrated robot kinematics and distance constraints with obstacles into the optimal MPC controller for single robot [50], [52], and robot swarms [38]. Fuzzy logic control (FLC) provides another intelligent decision way with the fuzzy set, which can be used to describe the ambiguous variables and draw a conclusion similar to human thought process. This kind of method has been successfully integrated into the shared control of 1) ERD/ERS-based robots with a fuzzy discrete event system to extend the robot behaviors with obstacle avoidance and wall following [106]; and 2) SSVEP-based robots to track user intention and guarantee the robot safety [171]. In addition, based on the context of surrounding environment, there is a strategy with virtual attractor zone for given BCI commands and repellor zone for obstacles [17], [82], [105], [189].

2) Continuous Control Strategy: Although most of current studies aforementioned are based on the discrete interaction modality, a continuous control strategy would rather be more desirable from the perspective of long-term development. The scheme of directly mapping the ongoing EEG to continuous movement signals, rather than through a decoded command from BCI, is the design ethos for a continuous control strategy. This is, however, very challenging due to the inherent non-stationary characteristics of the EEG signal, and may lead to unstable control performance of the wheeled robots. So far, according to our best knowledge, there is no related demonstrations for such an EEG-based wheeled robot system.

Instead, an idea was to translate decoded BCI outputs into a continuous signal for the terminal robots. Since the decoded output for an ERD/ERS-based BCI system can be viewed as a stream of continuous probability distribution rather a discrete command, Tonin et al. [113] proposed a novel control framework on the basis of the dynamical system which explicitly considered the previous system state and the current BCI output through a linear combination. Compared with the traditional discrete modality, the resultant system of a brain teleoperated robot proved to be more reliable.

C. Robot Intelligence

The robot intelligence for EEG-based systems is able to bridge the robot to the real-time environment, while providing some necessary information to the control strategy. Generally, the robot intelligence is composed of perception, planning, and motion tracking [17], [44].

Specifically, the perception of robot intelligence including awareness of the robotic states and the environments. For the former perception, a few simple sensors are sufficient to obtain the robot information of position, angle, and velocity. The most extensively used sensors for self-states perception include the encoders (see in [21] and [69]), gyroscopes [144], and inertial measurement unit (IMU) [89], [91]. On the other hand, the environmental perception is usually composed of three main modules of the situation awareness, obstacle detection, and simultaneous localization and mapping (SLAM). The task of the situation awareness module is to identify any environmental dynamics, semantic changes, and ambiguous situations that needs the user collaboration [17], [44], [73]. The common employed sensors for obstacle detection consist of range sensors, such as the infrared proximity sensor, ultrasonic sensors, touch sensors, radar sensors, and laser range finder (LRF). Several visual sensors as webcams or cameras have also been deployed in a numerous studies to facilitate the remote interactions [17], [89], [142], [160]. If the robot is operated in an unknown or uncertain environments, a SLAM algorithm is generally further incorporated [110], [111] (see in Table I).

The planning module serves as the core part of the robot intelligence, which handles the perceived information and provides the appropriate set of linear and angular velocity commands to the subsequent motion tracking module. The basic idea of this part is to search for an optimal or near-optimal collision-free path for an EEG-based robot [89]. The planning methods can be roughly classified into global, local, and hybrid planners. Global path planning is based on the knowledge of global map information to develop the optimal path from the starting point to the desired destination by sampling-based or search-based approaches. The search-based method, which generally produces a graphical representation of the planning issue and finds the shortest path in a graph, is more preferred in the EEG-based robot system with algorithms such as Dijkstra [85], [167], and A* [160], [199], [206]. On the other hand, a local planner works in a completely or partially unknown environment, which are detected online by sensors to obtain the information of location, shape, and size of obstacles around the robot. The main local planning algorithms include artificial potential field (APF), vector field histogram (VFH), and the dynamic window approach (DWA). The basic idea of the APF is to consider the robot movement in the environment as a virtual motion in an artificial force field, which has been widely used in generating a smooth and collision-free path for EEG signals [130], [131], [132], [169]. Unlike the APF, the DWA approach incorporates the robot dynamics and searches for the controlling commands directly in the space where the velocities are safe with respect to the obstacles. The method has also been successfully used in a mentally driven telepresence robot [167]. There are also some local planners based on other approaches, such as D*Lite algorithms [21], the Frontier point method [5], being used to calculate the motion direction of brain-controlled robots. A hybrid motion planning architecture has also been applied in the EEG-based robotic systems [44], [73], [120], which integrated a fast three dimensional global path planner (consisted of the A* approach, an interpolation module, and a smoothing algorithm) and a local planner based on the double DWA method. The related findings of real BCW experiments have shown the effectiveness of this hybrid planner.

The main task of the motion tracking is how to regulate the actual velocities to match the planned desired velocities. Generally, proportional-integral-derivative (PID) controller is integrated for this purpose by default in off-the-shelf vehicles, wheelchairs or the mobile robots [5], [18], [47], [122], [160], which continuously calculates the error between the desired setpoint and the current value of the velocity, and attempts to minimize such error over time via the classical modulating terms of a proportional, integral, and derivative part. An adaptive control algorithm that varies from PID has been proposed in an actuation system to regulate the motion of an SSVEP-based BCW [127]. The controller considers the wheel dynamics within the design of the control law, and finally generates the pulse width modulation (PWM) signals to the motor driver. Moreover, considering the impact of rapidly changing brain-controlled commands to the EEG-based system, some studies have investigated a more robust velocity servo control. Sliding mode control (SMC) is regarded as a powerful robust control approach, which is insensitive to the parameter perturbations and external disturbances. Li et al. worked on the integration of SMC with EEG-based robots. The experimental findings with developed sliding mode controller showed that the proposed approach was feasible [51], [52], [134].

IV. Performance Evaluation

By leveraging the system architecture and key technologies, a brain-controlled wheeled robot can be rationally established. After that, how to judge the system performance in terms of a specific task is important. The evaluation of such fused system is rather complex, with many factors, such as the participant, experimental protocol, tasks and environments, and evaluation matrices, that are critical and need to be carefully considered for the overall system performance [211]. Here we summarize the general aspects of the above mentioned factors. The specific examples of each part are listed in Table III, and the detailed descriptions are as follows.

A. Participants

Most of the current successful demonstrations have been conducted with a selection of certain participants, where defined inclusion and exclusion criteria are applied. Generally, a consensus is that the exclusion criteria should include the following aspects: 1) individuals with cognitive deficits that prevent to remain concentrated during the whole experimental session; 2) people using any type of psychiatric medication; 3) users who experience hallucinations, dyslexia, or episodes of epilepsy; 4) other conditions that rendered the individual unfit to take part in the study, such as pregnant women, history of visual defects for the visual evoked EEG paradigms, etc.

The inclusion criteria, on the other hand, are not rigorously consistent in different studies. The total number of the involved participants generally varies from one to twenty, while several studies have recruited larger numbers of subjects such as 50 in [195], 61 in [129]. Besides, there are differences in the health status, the average age, and gender, as listed in Table III. Regarding the health status of the participants, most EEG-based robotic systems' experiments have been carried out with mere healthy participants. A few investigations attempted to assess the system performance only using volunteers with motor disabilities, while a fair number of research groups recruited both the healthy and disabled volunteers to evaluate the system performance. In the majority of these cases, the number of healthy users is usually greater than that of the disabled.

Regarding the average age, interestingly, young users make up the vast majority of the participants among the existing studies. In detail, the users who participated in the experiments of EEG-based wheeled robots were either generally described as "young", or in the specific age range between eighteen and thirty years. Some scholars even define the age in the inclusion criteria (e.g., the requirement of 20∼25 years of age [21], [72]). This may be explained by the facts that participants are usually recruited from university students [56], or the future-oriented brain-controlled intelligent systems are more attractive to young populations [195]. As for the gender, the number of male subjects is always greater than, or at least equal to that of female with several exceptions such as in [35], [194], and [199]. In fact, Li et al. found that the age and gender of the participant seem to have an impact on the ability to recognize EEG signals [110]. However, since the influence in their data was slight, no further investigations have been explored, which we regard deserves to be instigated by more samples in the future.

B. Experiment Protocol

An experimental protocol is indispensable for the evaluation process of the system. In general, the objectives and the procedure of the experiments are explained to participants in advance. To preserve the health and rights of the participants and to follow the ethical standards, all processes of the research need to adhere to the Declaration of Helsinki. Some of the experimental protocols were further approved to be responsible for the study by the local Ethic Committee of the research institution (see in Table III). Meanwhile, a signed informed consent document is usually required from all subjects. Prior to the experiment, the brain-controlled protocol does not require users to have experience with BCI/BMI, but some studies were conducted to involve relevant subjects (e.g., [14], [58], [112]).

The experimental procedures of numerous studies are broadly composed of two parts: an offline calibration/training session and an online test. The former stage is often included to calibrate the subject-specific parameters of BCI/BMI classifier. In general, this session consists of a set of predefined commands. In some cases, subjects have to reach minimum accuracy for the subsequent online test (e.g., [109], [110], [130]). In this case, the training session is repeated until the accuracy requirement is satisfied. The duration of these two stages is determined by the employed BCI paradigm, task difficulty, and specific accuracy requirements. Specifically, some studies may end quickly within hours or last a long time to several months (e.g., [45], [113], [121]).

C. Tasks and Environments

Brain-controlled robots aim to improve the users' mobility, hence the most widely adopted task is a BCI-based point-to-point navigation, in which individuals use their EEG signals to move the wheeled robots from a certain starting point to a destination point (e.g., [5], [11], [68]). As listed in Table III, in some online studies, the participants are required to operate the robots along a predefined route, while others, especially in an open space, do not restrict the users to a specific trajectory. In general, subjects are requested to finish the task as fast as possible to reduce the user workload. Often, subjects are also asked to avoid obstacles to ensure safety. Moreover, for the given tasks, limited speed of the linear and turning velocity is usually required to ensure the system safety. On this basis, an automatic-forward movement was always set as the default behavior of the controlled wheeled robots, with a constant forward speed. A task is generally deemed to be a failure while the robot could not reach the destination within the predefined time duration (e.g., [106], [133]).

The environments or navigation field can be either simulated or realistic. The simulated conditions have advantages

TABLE III

PERFORMANCE EVALUATION OF EEG-BASED WHEELED ROBOTICS [HEAL: HEALTHY SUBJECTS; DIS: DISABLED SUBJECTS; M: MALE; FM: FEMALE]

Index	Category	Description	Examples
Participants	Health Status	Only tested with healthy users	2 Heal: [11]-[13], 5 Heal: [21], [69], [70], [72], [117], [136], [188], [198], 8 Heal: [46]-[48], 11 Heal: [133], [175], [202], 13 Heal: [113], 19 Heal: [42], and other numbers in [14], [16], [167], [179], [186], [203], [206]
		Only tested with disabled users	7 with different disabilities [5], 3 with severe tetraplegia [45], 1 with tetraplegia [54], 10 with severe spinal cord injury [57], 1 ALS patients [73]
		Both the healthy and disabled	4 Heal & 2 Dis [17], 7 Heal & 6 Dis [44], 4 Heal & 3 Dis [56], 5 Heal & 5 Tetraplegia [65], 10 Heal & 9 Dis [105], and others [120]-[122], [126], [132], [156], [169]
	Age	Young	[38], [68], [69], [117]
		Specific age range	Age between 15~34 [5], 23~28 [17], 21~36 [20], 18~27 [35], 22~34 [36], 20~35 [42], 22~36 [111], 25~30 [124], 24~32 [145], 22~27 [155], 18~20 [179]
	Gender	More males	5 M & 2 FM [5], 4M [15], 3 M & 1 FM [19], 12 M & 4 FM [20], 11 M & 1 FM [33], 8 M [46], 8 M & 2 FM [57], 13 M & 7 FM [63], 6 M & 1 FM [85], 6 M & 4 FM [106], 8 M & 5 FM [115], 11 M [133], 8 M & 2 FM [148]
		More females	3 M & 12 FM [35], 1 FM [167], 1 M & 2 FM [179], 3 M & 7 FM [194], 1 M & 3 FM [199]
		Equal number	3 M & 3 FM [58], 1 M & 1 FM [71], 2 M & 2FM [98], 5M & 5 FM [186]
Experiment Protocol	Protocol	Helsinki Declaration	All need to adhere
		Local committee	[35], [44], [56], [65], [86], [105], [115], [122], [133], [146], [184], [206], [210]
		Informed consent	[10], [33], [44], [45], [75], [113], [115], [121], [125], [129], [186], [203], [206]
	Procedure	Offline training time	5 days [12], 3 weeks [72], 9 days [105], one week [151], several months [45]
		Offline calibration & Online	[57], [72], [109], [110], [120], [130], [133], [134], [151]
Task and Environment	Tasks	Predefined route	[103], [109], [130], [133], [136]
		Free trajectory	[16], [52], [113], [134], [151]
		Requirement	Accomplish the tasks as quickly as possible [15], [45], [101], [105], [136], [151] Avoid obstacles [16], [52], [55], [72], [73], [106], [151]
		Speed limit	Linear speed limit for BCV: 1 or 1.5 m/s [19], 0~10m/s [47], 2~3m/s [20], [147], [162] for BCW/BCTM: 0.15~0.3 m/s [109], 0.1~0.3 m/s [151] Turning speed limit: $\pi/3$ rad/s [5], 0.15 rad/s [151], 42.9 degree/s [98], 10°~30° [146]
		Online setting	Constant forward speed of 0.15 m/s [52], 0.5 m/s [70], 0.2 m/s [113]
	Environment	Simulated	Simulated traffic scenario [58], Gazebo environment [89], virtual [179]
		Realistic	[33], [36], [37], [52], [109], [111], [130], [144], [167]
		Structured	For BCW/BCTM [14], [56], [68], [70], [109], [127], for BCV [20], [48], [147]
		Semi-structured	[15], [44], natural working space [105], a domestic room with static obstacles [160]
		Unstructured	Open region [52], [113], lift hall [68], and others in [16], [112]
Evaluation Metrics	BCI/BMI	Accuracy	Almost used in all relevant studies
		Response time	[56], [70], [122], [160], [175]
		ITR	15 bit/min for P300 [21], 26.57 bit/min for ERD/ERS [99], 60.45 bit/min for SSVEP [149]
		Errors	Total errors [21], [44], [72], error rate [70]
	Navigation	Task success rate	Average success rate [13], [16], [21], [44], [56]-[58], [73], [101], [120], [131]-[134], [141], [145], [147], percentage of reached targets [14], [115], number of task accomplished [72], [109], [111], [130], [151], [169]
		Task time	Task completion time [5], [11], [15], [33], [72]-[74], [134]-[136], [146], [151], [171], [189], [191], [202], ratio of operating times [11], [47], [105], time optimality ratio [19], [72], [112], [145], number of commands [11], [44], [115], [189]
		Trajectory	[5], [12], [17], [21], [33], [36]-[38], [111], [113], [128], [130]-[134], [141], [142], [175], [188], distance [12], [48], [73], [141], path length optimality ratio [45], [72], [112], [145], [146]
		Average speed	[21], [47], [133], [134], [169]
		Collision number	[35], [72], [120], [206]
	User	Methods	NASA TLX questionnaire [44], [63], [106], [112], [122], customized questionnaire [33], [52], [72], [113], [133]
		Workload	Average number of commands [11], [15], [44], [189]
		User states	vital signs [57], emotional state [44]
	Robustness	For environments	Different constraints [72], suddenly appeared persons [130], irregular dynamic obstacles [133]
		User adaptability	Ability to manage varied tasks [21], [47]
		Decoder robustness	Related to BCI accuracy [13], [18], [82], [115], [133]
		Motion robustness	Unexpected disturbances [73], [85], [127], [130], velocity regulation [51], [52], [134]

of safety, convenience, and low consumption. More importantly, by incorporating various robotic systems and creating experimental conditions, it provides a better choice under some dangerous scenarios to test the feasibility of related designs. Indeed, for the concern of users' safety, most current investigations on BCVs were conducted in a simulated scenario. However, it is worth noticing that even best simulated platforms cannot exactly replicate the actual physical system. It seems more reasonable to use both simulated and realistic environments to evaluate the system in different developing stages [52], [103]. Moreover, from the viewpoint of the robot movement, the environments can be also further classified into structured, semi-structured, and unstructured. A structured environment refers to a fixed layout of the surroundings where the robot works. Typically, several predefined home/office/road-like scenarios with static obstacles belong to this category. Similar to the early and typical developed BCWs and BCTMs, most current studies on BCVs use a virtual structured scenario to test the viability of developed approaches. The corridor is a semi-enclosed space consisting of a doorway and two-side walls, which has been a common tested scenario, such as in [21], [72], [82], and [167]. Unstructured environments have non-fixed, unknown, and indescriptive environmental information, which have not been reported to be employed in EEG-based mobile robots. In addition, there are also many studies focused on the semi-structured environments, where the environmental information is partly known (generally being a fixed indoor layout with unknown moving obstacles).

D. Evaluation Metrics

The evaluation criteria employed in different studies vary from each other greatly. Different tasks, environments, and metrics are used to evaluate the related systems. Beyond the most fundamental requirements of the safety and efficiency, several other performance issues have also been spotlighted, such as the behavior of the participants [211], [212]. Some scholars compared the system performance of manual control (e.g., with the joystick, buttons, or keyboards) with the BMI control, while others conducted a quantitative evaluation of the system controlled solely with BMI or the proposed assistive approaches. All these efforts result in a diverse benchmark and, as a consequence, a lack of unified gold standards.

Nevertheless, several common evaluation metrics have been employed to help analyze the performance of proposed system and facilitate the comparison with other systems. Overall, four performance aspects of BCI/BMI, navigation, user, and robustness are most highlighted in the current evaluation of EEG-based mobile robots, as listed in Table III.

1) BCI/BMI Performance Metrics: Many metrics are available to evaluate the effectiveness of the used or proposed BCIs. The most widely applied metric is the accuracy, which indicates the EEG recognition accuracy (ERA) of the decoding algorithms. The average BCI accuracy of the evoked signals (e.g., SSEP, P300) in many studies reached more than 90% [72], [85], [175], [206], higher than that of reported spontaneous signals. The response time, which is the interval from a command intended to be output to that it is actually issued, has also been applied in numerous research. Other metrics are related to errors, for instance, the total errors (e.g., wrong detected targets), or the error rate (i.e., the wrongly number divided by total number).

Indeed, an important index of the information transfer rate (ITR) has been extensively used to evaluate a BCI system, which is defined as follows:

$$B = \frac{60}{T}\left(\log_2 N + A\log_2 A + (1-A)\log_2\frac{1-A}{N-1}\right) \quad (1)$$

where B is the ITR in bit/min, N is the number of possible selections, and A is the accuracy, and T is the time required to output one BMI command [213].

From the above definition, the ITR depends on both the accuracy and speed of a BCI. Generally, the SSVEP has better ITR because of its relatively larger number of commands, faster response time, and higher accuracy. A caveat of these BCI measures is that, during the actual navigation, it is not always known which BCI command the user wishes to deliver as they may complete the task in different ways (see [45] for how to assess BCI performance in this situation).

Other metrics of BCI/BMI performance are applied to evaluate user interfaces, such as the concentration time [160], the command utility (i.e., command usage frequency) [21], [72], and the selections per minute/mission [21].

2) Navigation Performance Metrics: Certain metrics have been widely used to evaluate the navigation performance of an intelligent mobility device [214]. One is the task success metric, which indicates the condition of the completion for a given navigation task. The most widely used index is the task success rate, defining as the number of successful trials over the total conducted trials, and the value can reach up to 100% [56], [131]. It is worth noting that in those studies with fewer trials, the number of accomplished tasks is directly used. Moreover, in [57], a related metric of navigation error rate was used to count the number of task failures caused by the error navigations. The second navigation metric is related to the task time, such as the task completion time (i.e., the total time taken to finish the task), the best navigation time [82], the ratio of operating times (i.e., manual time divided by the mental time), and time optimality ratio (the ratio of the time spent to the optimal time). The number of commands is also a time-relevant metric as the majority of brain-controlled devices set the BCI response time fixed. Similar to the completion time related ratio, the command ratio is also used [122], [167].

Trajectory is an important metric for qualitative analysis of the navigation performance. For the quantitative evaluation, the distance or path length of the followed trajectories has been applied. Further, the path length optimality ratio of the traveled distance to the optimal path length was used. For the systems with predefined routes, the number of missed waypoints is counted [109]. Moreover, given the traveled distance and task time, the average robot velocity can be further derived to indicate the path efficiency.

Another important metric is the collision number (either with the wall or the obstacles) during the task. In addition, for the more complicated BCVs, there are some other metrics to evaluate the navigation performance, such as the mean lateral error, the out-of-boundary rate per kilometer when steered in a structured road [19], [147], and the inter-vehicle distance [47], [48], tracking error index (TEI) [48] of a car-following task.

3) User Metrics: Here, we outlined the most commonly used metrics for the EEG-based wheeled robotics to assess the ergonomics (which generally implies the user comfort) and user activity (related to the interaction with the robots). The focused metrics are usually measured by a questionnaire, which come in two different formats. The first follows a specific standard, such as the extensively used National Aeronautics and Space Administration-Task Load Index (NASA-TLX). Besides, a customized questionnaire with questions directed to the case-specific tasks have been used widely. In 2021, both of these two methods have been used to completely assess the user experience for the developed BCW system, which providing an important reference of the field [44].

Workload is one of the most important ergonomic metrics, representing user's level of effort during the brain-controlled tasks. The average number of commands has often been used as quantitative index to evaluate the workload. Moreover, several studies attempted to obtain the learnability and level of confidence from the questionnaires. The learnability relates to the easiness to learn how to use the system for the assigned tasks, while the level of confidence refers to the users' confidence when performing the tasks. The metrics of workload, learnability, and the level of confidence have been successfully used for the psychological assessment [21], [72]. Furthermore, some indexes monitor the states of participants, such as the vital signs (e.g., blood pressure, heart rate, and respiratory rate) [57], and psychometric questionnaires for user emotional state [44].

On the other hand, the participants' interaction strategy when controlling the robots via EEG signals is investigated by the activity analysis. To evaluate the direct-control-oriented and shared control supervisory-oriented interactions, Escolano et al. defined several specific metrics, such as the discriminant activity, the control activity descriptor, and the supervisory activity descriptor, which quantify the ratio of the focused selections to the entire number of options [21], [72]. In addition, the level of assistance of the control intelligence, especially in a semi-autonomous control, is assessed in relevant studies [115].

4) Robustness Metrics: Robustness is defined in various ways, but all reflect the degree of how well a system performs when faced with unexpected inputs [214], [215]. In other words, robustness mainly refers to the adaptation ability to the unpredictable fluctuations. In studies of EEG-based wheeled robots, such ability can be measured in four ways: adaptability to environments, user adaptability, decoder robustness, and motion capacity to disturbances (see in Table III).

For the adaptability to environments, many researchers paid attention to the system capability to handle the various obstacles, which include both static and dynamic ones. Taking the BCW system as an example, the experiments in semi-structured environments have been conducted considering factors of suddenly appeared persons [130], irregular dynamic obstacles [133], and different constraints [72]. Generally, the evaluation metrics of such robustness are related to navigation ones, such as the success rate, collisions, average velocity. As for the user adaptability, it measures the participants' ability to manage varied tasks [21], [47]. However, no direct metrics are found to assess this ability, while generally being reflected by the metrics of task success (navigation metrics) and ergonomics (user metrics). Another important aspect of the decoder robustness, which is related to the BCI, has been widely studied. A large number of specific methods, such as the sliding window [47], [48], shared intelligence [115], deep learning approach [133], and majority weight of the analysis algorithms [83], have been proposed to improve the accuracy of the detected results. Meanwhile, the combination with the other biological signals (such as EMG in [142] and [144]) has also been investigated to enhance the robustness of overall system. In these cases, the metric of accuracy (BCI metrics) is mainly used.

On the other hand, an increasing number of researchers are interested in the robustness of the controlled robots, especially their robustness to disturbances and perturbations. Several studies have been conducted to investigate the feasibility to resist the changes of load, surface roughness, and other unexpected disturbances. Metrics employed to evaluate the effectiveness of such proposed methods are mainly related to the navigation performance, accompanying with some novel indicators. To examine the robustness, Li et al. calculated the regions encompassed by the actual and intended velocity profiles [51], [52], [134]. Also to quantify the robustness via the velocity curves, the smoothness of the related maneuver were recorded by the number of movements and computed as the inverse of the velocity profile's number of peaks [216].

V. Discussion

Brain-computer interfaces, circumventing the traditional neuro-muscular pathways, provide human an alternative approach to operate external devices, which is a promising technology. The integration of EEG-based BCIs with wheeled robots has drawn an explosive growing interest for assistive mobility purpose. As summarized in previous sections, great efforts in various aspects of these intelligent systems have been made. Nevertheless, to bring such the next-generation products from the laboratory to real scenarios, there are still room for further improvements and breakthroughs in both the system development and evaluation. In the sequel, we discuss several key open challenges and potential future research directions.

A. Open Challenges of the System Development

1) Natural and Efficient Paradigms: The first open challenge of EEG-based wheeled robotics relates to the choice of natural and efficient BMI paradigms. The pitfalls in the BMIs design has been illustrated in detail in [212]. Current BMIs are, in fact, not actually designed for the brain-controlled dynamic systems, although they have been directly transferred from the traditional communication domain to robots. Unlike those letters selection or cursor movement on the screen, EEG-based wheeled robotics generally have more sophisticated control requirements due to their increased degrees of freedom (DOF) and specific tasks. For example, there are at least six available commands for robot start/stop, turning left/right, and speed control of acceleration and deceleration [109]. Taking into account the time-sensitive, stringent safety, and reliability requirements of such systems, especially for those manned robots, failure to issue the right command at the exact right time could result in serious injury. In addition, such applications are also typically required to operate in highly dynamic environments, with variations in the surrounding objects during the task performing. Thus, to ensure that the mechanical chassis with dynamic inertia can be naturally and safely applied in practice, some combinations of basic motions, such as forward-steering, mode switching, and even turning under varying speed or controlling of the integrated robotic arms during the motion will need to be attained with fine-grained control, which poses a great challenge to the BMI paradigms used. Although several mainstream paradigms are able to achieve these basic control as technology advances, a natural and efficient paradigm remains a great challenge. Here, we mean "natural" in the sense that the paradigm can smoothly induce the appropriate control signals according to the users' intention and be widely accepted by the end-users [8], [113]. Moreover, "efficient" refers to that the paradigm can be successfully employed with minimal training, less mental loads, and good performance [195].

Since users may modulate their brain signals by volitionally engaging in mental processes, and the efficacy as well as acceptability have been repeatedly proven [10], [29], present MI paradigm appears to be more natural. However, the lengthy user's training time of such modality have restricted its practicability concerning the long-term consistency and stability. Moreover, the related decoding performance (e.g., ITR) is still insufficient. Exogenous stimulation-based paradigms, on the other hand, have shorter training periods and higher recognition accuracy, making them more efficient [20], [206]. Nonetheless, the presence of stimuli may cause a shift in user focus, in particular, the most commonly used visual stimuli can easily fatigue users. Besides, the predefined relationship between the stimuli and robot actions narrows the full free brain-control of the system, which seems not to be natural enough (i.e., lacks the smooth feeling) under daily use. Hence, it is still an open challenge to design a natural and efficient paradigm, able to distinguish high-dimensional commands, that enables users to control the wheeled robots via brain signals continuously and effectively.

2) End-User and Robot Coupling: A natural and efficient paradigm is the paramount promise for EEG-based wheeled robots in the long term. However, given current paradigms, how to improve the coupling between the user and robots is a more urgent and fundamental challenge. In most studies, the terminal robots are only considered as the passive actuator of a BMI [29], raising two main problems in the human-robot symbiotic interaction.

Firstly, the characteristics of the robot itself are not fully considered in the design loop from the human to robots, thus the performance of whole brain-controlled system is affected. Over the last two decades, the research priority has gradually shifted from the BMI component (i.e., mainly the algorithms for intent decoding) towards the control strategies. In particular, the emergence and flourishing of the shared control, through the fusion of user and robot intelligence, has provided necessary guarantees of the safety and navigation performance for the brain-controlled systems. However, due to the general "either/or" approach, none of the current shared strategy can truly reflect the synergy between the human and robots. Therefore, more flexible fusion frameworks remain to be explored to achieve a higher level of human-robot hybrid intelligence. Furthermore, little work has been done to deeply investigate the influences of the delivered BMI signals for the robot itself, especially for the robot motion tracking of dynamic velocity regulation within overall neuro-robot characteristics considered. This is rather important for a stable and robust real application, especially considering that the terminal wheeled robots are distinctive to be restricted by the instant brain commands and their complicated dynamics, such as the nonholonomic and internal constraints of velocity increments, actuator dynamics, and other saturation limits [52], [130], [134]. The robust velocity modulation, safety, and navigation performance are all critical for EEG-based robots, and the robot's inherent capabilities should be adequately leveraged to strengthen the interactions between BMIs and robot itself.

On the other hand, from the perspective of a closed-loop system, research in mutual learning between the users, BMI, and robots is limited. There is a widespread faith in the field that coadaptation between the user and brain-actuated robots is a direct and automatic result. It is important to highlight the fact that such interaction not only improves user operational efficiency, but also facilitates the acquisition of BMI skills [29], [45], [113]. Longitudinal training of the users is one of the effective ways to study the effects of learning. For instance, Herweg et al. [78] conducted tactile ERP-based BCW experiments with five training sessions in a virtual navigation task, and showed the positive effect on improved BCI performance of accuracy and ITR. An indoor simulated environment for SSVEP-based BCW was designed in [206], and the seven-days training in different situations showed an increase of the decoding accuracy and commands. Most recently, Tonin et al. investigated the learning phenomenon between the user (i.e., adaptation to the BMI decoder) and the BMI (i.e., concurrent re-calibration) [45]. The results of three tetraplegic users for the MI-BCW over several months exhibited a better navigation performance. Despite these cases, the explicit experimental evidence of such mutual learning among the agents involved in the EEG-based wheeled robotics is still scarce [217]. In particular, for the systems relying on a large number of mental commands in complicated scenarios, how to couple the mutual learning remains unclear and needs to be further explored.

B. Open Challenges for the Standardized System Evaluation

Another limitation involves the issue of how to perform a standardized system evaluation of EEG-based wheeled robots despite their different technologies. Rebsamen et al. [70] presented a cost function to quantify the control efficiency in order to assess the overall performance of the developed BCW and compare it with other P300 and MI based systems described in the literature. The cost was computed by adding the relevant ratio of the mission time and concentration time of BCI, which were normalized by the nominal values. Noting that the nominal time is defined as the required minimal time to reach the destination. Bi et al. further extended the comparison results with several SSVEP-based robots [22]. However, such function cannot reflect the overall system performance, because some vital performance aspects, such as the BCI, safety, and robustness, are not included. Meanwhile, to make the focused systems comparable, several hypotheses that do not correspond to reality were often made, neglecting the conditions of varied environments, tasks, and the health status of the participants. Based on the principle that the task should be simple for fewer classification mistakes while the number of commands is supposed to be maximized for more control freedom, the command to task ratio (CTR) between the number of available commands and tasks was used to evaluate varied EEG-based robots [23]. The ratio indicates how easily additional commands can be introduced. However, it is not always appropriate to compare systems based on different BCI paradigms because the targeted tasks involve different brain cognitive processes.

Moreover, besides the metrics mentioned in Section IV-D, there are also several critical requirements for a EEG-based wheeled robotic system to land as practical applications. The usability of the device has been highlighted in a number of studies [45], and is generally evaluated by the users through a questionnaire. Due to individual differences and various scoring scales, it is difficult to properly view the obtained results of each developed system. More objective metrics are required, including the lightness of recording device (which lessens the burden on the head), system's preparation time, stability, and ethical safety, etc. Indeed, considering that the overall system performance involves numerous metrics with different levels of importance, how to establish a universal evaluation criteria will be the forthcoming research issue of future work.

C. Future Work

The non-invasive brain controlled wheeled system for single user or single robot is becoming increasing mature, while multi-users and multi-robots teams are just emerging and will be the long-term goal of the community. However, the open challenges abovementioned have hindered the transition of BCI from the lab to clinics and end-user homes, and future research could continue along three axes.

1) Towards BCI Modalities: The current decoding of EEG signals presents some difficulties, including the poor spatial resolution, low signal-to-noise ratio, and a restricted number of commands. Thus, one important research direction is to further improve the performance of BCIs by decreasing the average response time, increasing decoding accuracy, and broadening the coverage of system's brain-controlled signals. Specifically, for the online artifact removal, feature extraction, and classification, there are still traditional methods being widely used for the real-time EEG-based robots. Powerful artificial intelligence (AI) algorithms, such as the deep learning [218], has been considered for the online use. Although it is difficult, in the real world, to collect a large enough datasets to ensure the decoding accuracy because of the real-time requirements, site constraints, and environmental impacts, the idea of transfer learning can be adopted [219]. Specifically, one can create the complete operational data in a virtual environment and then migrate the control experience to the actual world.

On the other hand, incorporating more human users' sensory modality into BCIs, such as the olfaction [220], [221], [222], auditory [223], [224], tactile P300 [225], visual imagery (spontaneous mental simulation of visual information from memory) [226], may be a powerful way to neutrally modulate the wheeled device control. To seek more natural and efficient paradigms, another potential approach is to develop hybrid BCI paradigms [227]. For instance, since the functional NIRs is less susceptible to artifacts and can be portable, it can be coupled together with EEG for enhanced performance [228], [229]. Moreover, instead of the existing concept of distinct paradigms controlling different DOFs, more dexterous paradigm would be an essential premise to address the as-described challenges. For example, the user could implement fine robot motion control through the natural MI paradigm, while managing discrete events via evoked paradigms [29].

2) Towards Human-Robot Interaction: Although improving the BCI performance is desirable for future work, we argue that delivering the proper commands at the correct time is more important for efficient mental control of an EEG-based wheeled robot [15], [45], [52], [105]. This actually reflects the user's brain-control skills and the quality of interactions between the user and robots. Thus, there is a continuing demand for developing better shared human-robot control approaches [230], including intelligently assigning the control weights to users and robots in order to optimize the system performance. Detailed research can be carried out from both the human and robot sides. First, the consideration of the human impact and varied human responsibilities in designing a more flexible shared framework is an essential avenue for the future study. Second, the open-source ROS has a tremendous modularity and expanding community, which offers a common approach and message format for the traditional human-robot interaction. Hence, integrating it with BCIs for bidirectional communication would improve the control efficacy of the ecosystem, and has been attempted in EEG-based robots [56], [91], [167], [168]. This might be an important area for future research.

Besides, although the technical integration will lead to the natural modularity of users, BCIs, and robotics, the interaction characteristics of coupling such an ecosystem needs to be better understood. To have a deeper understanding of the whole EEG-based wheeled robotic systems, one could provide more intuitive and fluent interaction approaches, such as the adaptive interfaces based on goal recognition [210], or develop brain-controlled operator models with high fidelity and integrate them with the mature dynamic models of wheeled robots, as described in Section II-C2.

3) Towards a Unified Evaluation Architecture: Systematic evaluation covers many factors such as subjects, experimental protocols, tasks and environments, and evaluation metrics. In future research, more studies are needed to verify whether the good performance achieved with existing selected subjects is equally applicable to all the targeted population (including different health conditions, age distribution, gender, etc.). To compare different systems, a unified architecture with standard experimental protocol, tasks and environment, and evaluation metrics is needed. Control capability is a desirable unified criterion to measure the ability of a BCI to translate neural activity into control signals, and to perform complicated control operations. To test such a capability, a metric of Fitts throughput, defined as the ratio of the difficulty factor to the time required to move to the target position, has been used in

BCI research [231], [232]. We expect that such quantitative and comprehensive metrics can be introduced into EEG-based wheeled robotics in the future.

VI. Conclusion

BCIs provide a novel and feasible way to convey the users' intentions to the terminal devices through modulation of brain signals. EEG-based wheeled robotics promise to improve human daily life with enhanced level of mobility, and have been vigorously developed over the past two decades. This survey paper presents a comprehensive literature review on this field and its components. We first introduced the fundamental information about EEG-based wheeled robots, describing the general system architecture, typical system types, and the main research efforts on the whole-system design. From the perspective of relevant key techniques, we summarized the ongoing progress of the related BCIs, control strategies, and robot intelligence. To evaluate the system performance, we also presented the conditions of the enrolled participants, the experimental protocol, the tasks and environments, together with the commonly utilized metrics of the performance with respect to the BMI, navigation, user, and robustness. The review ends with a discussion of several open research challenges and directions for future work. We hope that the paper will help those interested in such area to have a systematic, thorough, and better understanding of the related research. Although a number of technical gaps need to be bridged to better assist the human users, we believe that a practical EEG-based wheeled mobile robot will enter our daily life in the near future.

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